Power Outage Forecasting and Backup Generator Pre-positioning

95-828 Machine Learning for Problem Solving — Spring 2026

1 Introduction

Extreme-weather-driven power outages are a recurring societal cost in the US Midwest. We are given hourly outage counts and 109 atmospheric/hydrological features for the 83 Michigan counties over a 90-day window (**2023-04-01 to 2023-06-30**), and asked to (i) forecast outages over the next 24 and 48 hours and (ii) recommend the placement of five 1,000-household backup generators. Two design constraints shape the entire pipeline:

- **No future weather is available at test time.** Only outage history plus weather observed up to and including the forecast origin can be used.
- **The training window ends mid-storm.** The largest event of the year (peak 86,313 outages on 2023-06-29) starts on 2023-06-23 and runs through the end of training. The test horizon begins one hour after the final training observation, so any model has to extrapolate *into* a storm without ground-truth weather.

These constraints directly motivate our validation design (§3.1) and feature engineering (§3.2).

2 Data Analysis

The data is a single `.nc` file with shape T=2,161 × L=83 × F=109 plus an outage matrix $\mathbf{Y} \in \mathbb{R}_{\geq 0}^{T \times L}$ and a per-county `tracked` household count.

Sparse, heavy-tailed outage counts. 70.5% of (c, t) cells have zero outages, and the non-zero distribution is heavy-tailed (Fig. 2a). Across the full window, the state-wide hourly outage total ranges from 105 to 86,313, with the 99.5th percentile already ten times the median — a **zero-inflated, log-normal-tailed regression target**.

Storm-event structure. Using a state-wide threshold of 2,000 simultaneous outages with a 6-hour minimum duration (12-hour merge gap), we detect **26 distinct storm events** in the 90-day window (Fig. 1). The largest event (id 25, 2023-06-23 – 2023-06-30) has a peak of 86,313, more than 3× the second-largest (id 0, 27,643). The training window ends inside this event.

Strong autocorrelation. The state-wide outage series is highly autocorrelated, with a clear diurnal echo at lag 24h and a weaker weekly echo at lag 168h (Fig. 2b). This implies that lagged outage counts should dominate any feature set; weather is informative but secondary because its predictive value is partly already encoded in past outages.

County heterogeneity. Per-county `tracked` (the maximum number of households tracked) ranges from ∼5,000 to over 920,000. Wayne County (FIPS 26163, Detroit) peaks at 922,425 households tracked but its peak hourly outage of 15,143 is only the third largest, while Genesee (26049) and Macomb (26099) take disproportionate hits per household (Fig. 3a). This heterogeneity later drives our policy recommendation (§5).

Top weather correlates. The 15 weather features most correlated with state-wide outage are all moisture-and-wind-related (`cape`, low-level u/v winds, precipitable water, `vstm` storm-motion magnitude). We retain all non-zero-variance features (91 after filtering); tree models implicitly select among them.

3 Methods

3.1 Validation: Rolling-Origin CV that Mirrors the Test Setup

A naive last-48h holdout falls inside the largest storm of the year, so any single-window val-RMSE is a sample of one and dominated by storm-period error. We replace it with a **rolling-origin cross-validation** scheme (Fig. 4) consisting of 6 folds spaced 72 hours apart.

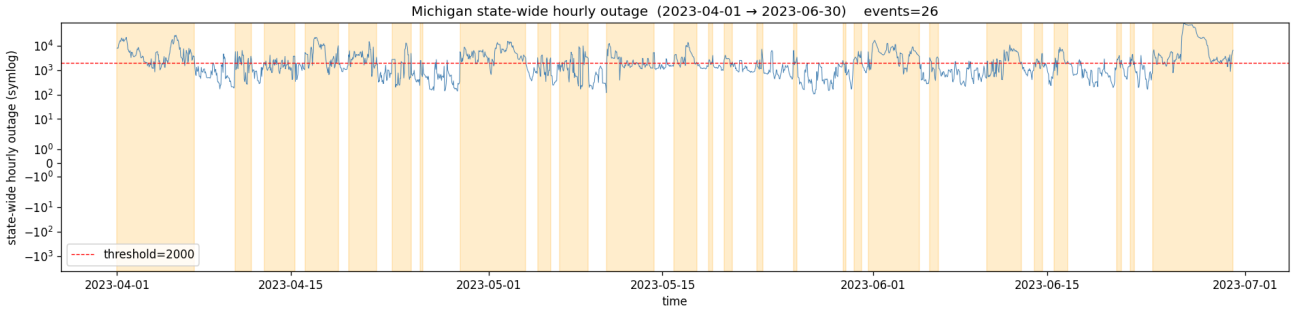
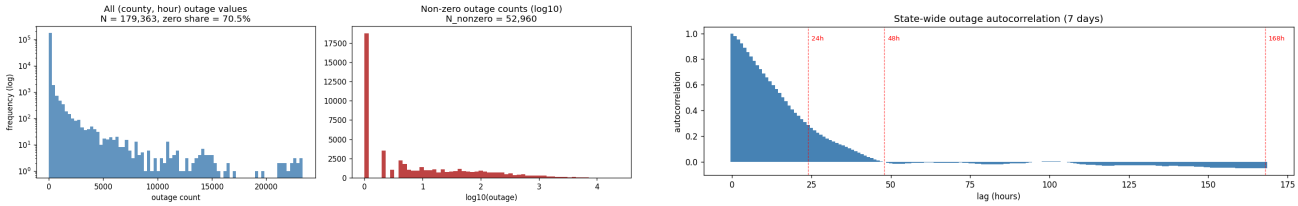


Figure 1: State-wide hourly outage total over the 90-day training window on a symmetric-log scale. Orange shading marks the 26 detected storm events; the red dashed line is the 2,000-outage detection threshold. Note the disproportionately large terminal event ending exactly at the train/test boundary.



(a) All-cell vs. non-zero cell distributions.

(b) State-wide autocorrelation (7 days).

Figure 2: Outage values are zero-inflated and heavy-tailed (left); the state-wide series exhibits strong diurnal autocorrelation (right).

- Each fold's validation window equals the forecast horizon (24 or 48 hours) *immediately following* its fit window — exactly the production setup.
- No weather observations from inside any val window are visible to its model.
- Folds span 2023-06-13 through 2023-06-29; one of them (fold 1) lands in the big-storm peak. Reporting mean±std across folds prevents a single high-leverage fold from determining the model ranking.

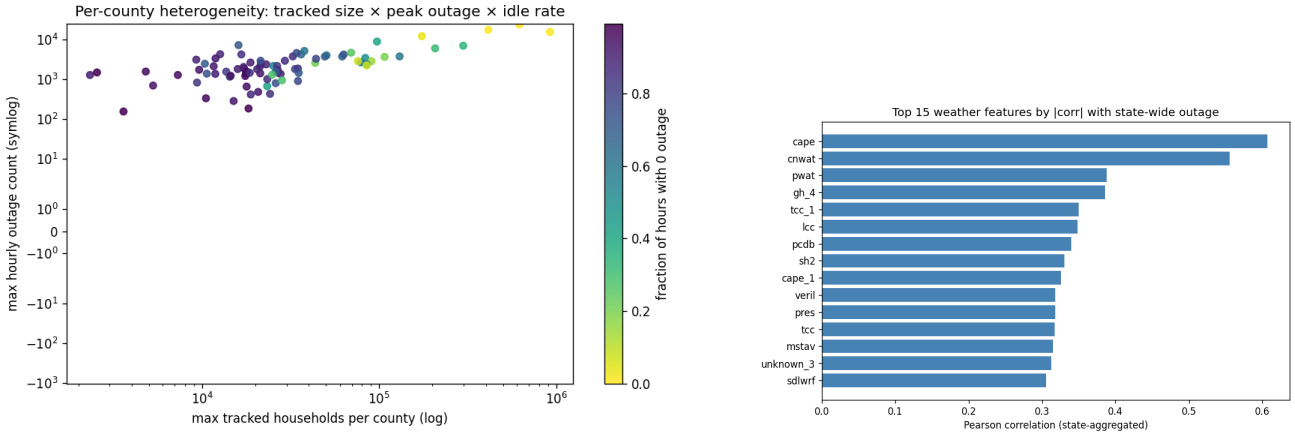
3.2 Feature Engineering

For tree-based and linear models we materialize a $(N \times P)$ feature matrix in which each row is one (origin time, county, horizon-step) triplet. Features (all leak-free, using only data up to time t):

- **Outage lags** — $y_{c,t-k}$ for $k \in \{1, 2, 3, 6, 12, 24, 48, 168\}$.
- **Outage rolling stats** — mean and std over $\{6, 24, 168\}$-hour windows.
- **Storm indicators** — rolling 24h max (peak intensity), 24h non-zero count (storm duration), $y_{c,t} - y_{c,t-24}$ (day-over-day delta).
- **Weather at t and lags $\{3, 12, 24\}$h** — captures pre-storm build-up.
- **Calendar at target time $t + h$** — sin / cos of hour-of-day, day-of-week, month-of-year (6 features).
- **Horizon step h** — so a single global model can output all 48 steps.
- **Location index** — ordinal for LightGBM/XGBoost, native categorical for CatBoost.

3.3 Model Families

We benchmark a deliberately broad set of models so that the report can defend each modeling choice with a baseline.



(a) County heterogeneity. Color = idle-rate.

(b) Top 15 weather features by |Pearson|.

Figure 3: Per-county heterogeneity and most-correlated weather features.

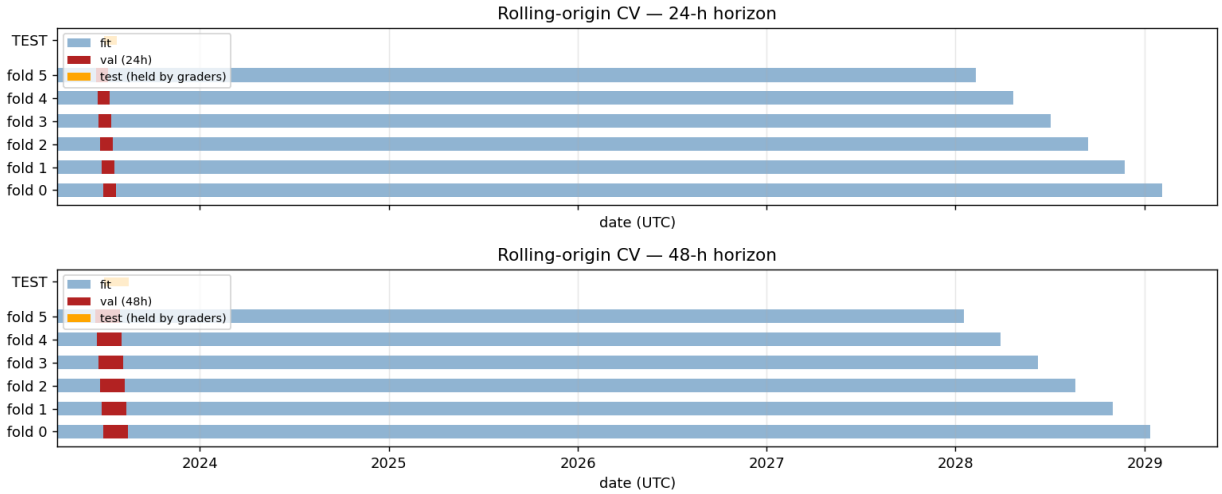


Figure 4: Rolling-origin CV layout. Blue = fit set (expanding-window), red = held-out validation, orange = the actual test window held by the graders. Each fold's val window matches the production constraint exactly.

- **Naive baselines (deterministic)**: all-zero, persistence ($\hat{y}_{t+h} = y_t$), seasonal-naive-24h, seasonal-naive-168h, and a per-(county,hour-of-day,day-of-week) historical mean.
- **Classical TS**: per-county SARIMAX(1,0,1), the second mandatory baseline named in the project description.
- **Tabular gradient-boosted trees**: LightGBM (log-target), LightGBM with Tweedie loss (compound Poisson–Gamma, designed for zero-inflated positive data), XGBoost, CatBoost (native categorical FIPS handling).
- **Hurdle/zero-inflated decomposition**: a binary classifier predicts $\Pr[y > 0]$; a regressor trained on positive rows only predicts the magnitude; the combined estimator is $\hat{y} = \Pr[y > 0] \cdot \hat{y}_{>0}$.
- **Per-county LightGBM**: 83 independent gradient-boosted trees, one per county, exploiting heterogeneity.
- **Sequence models**: GRU, BiLSTM, DLinear, PatchTST (run on H200).
- **Equal-weight ensemble**: arithmetic mean of the top-k CV models. We deliberately avoid *learned* stacking weights because all weight training would happen on the same val folds and would over-fit the big-storm fold.

Table 1: 24-hour rolling-origin CV: per-model RMSE (mean ± std across 6 folds, one seed). Lower is better. Fold count n may be < 6 if a configuration was still in progress at writing time.

Model	mean	std	min	max	n
Persistence (y_t repeated)	147.5	251.9	22.7	660.3	6
LightGBM (log target)	172.8	321.5	18.6	828.2	6
SARIMAX(1,0,1) per county	179.9	336.8	19.5	866.6	6
LightGBM (Tweedie loss)	182.1	342.9	19.6	881.1	6
All-zero	189.5	354.5	21.8	912.3	6
Linear regression on lag features	189.7	354.2	22.2	911.9	6
Seasonal naive (168h)	205.1	345.2	42.9	908.4	6
Hist. mean (county×hour×dow)	205.3	336.6	52.1	891.8	6
Seasonal naive (24h)	210.6	368.0	44.0	961.2	6
Ridge (α=1)	321.9	485.0	20.7	1207.0	6

Table 2: 48-hour rolling-origin CV: per-model RMSE (mean ± std across 6 folds, one seed). Lower is better. Fold count n may be < 6 if a configuration was still in progress at writing time.

Model	mean	std	min	max	n
SARIMAX(1,0,1) per county	157.9	274.2	33.9	717.1	6
LightGBM (log target)	158.8	274.6	33.2	718.6	6
All-zero	160.9	275.6	34.0	723.0	6
Linear regression on lag features	161.0	275.4	34.1	722.6	6
Persistence (y_t repeated)	178.7	286.1	33.5	759.8	6
Hist. mean (county×hour×dow)	179.5	261.7	64.2	713.3	6
Seasonal naive (168h)	183.2	263.2	51.5	719.8	6
Seasonal naive (24h)	217.9	270.3	35.9	737.6	6
LightGBM (Tweedie loss)	393.9	457.5	70.4	717.4	2
Ridge (α=1)	1712.5	0.0	1712.5	1712.5	1

4 Results

Rolling-origin CV summary. Table 1 reports CV RMSE for the 24h horizon; Table 2 the 48h. Both tables show the mean±std across 6 folds (one seed per stochastic model in this iteration). Two patterns are immediately visible:

(1) The mean is dominated by fold 1 (the storm peak.) For every model the per-fold maximum RMSE (achieved on fold 1) is one to two orders of magnitude larger than the minimum (fold 4, a quiet 48h window). The fold-level heatmap (Fig. 5) makes this concrete. This is exactly the symptom we expected from an OOD-storm test setup, and it explains why simple baselines like *persistence* score very competitively on mean-RMSE.

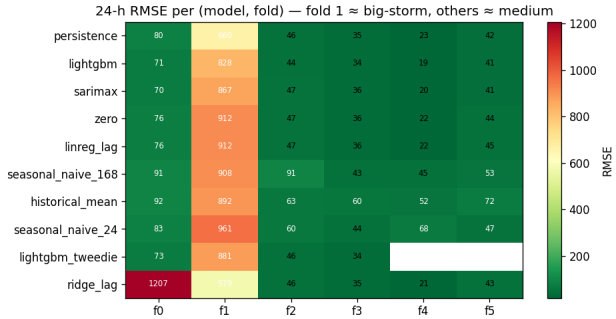
(2) LightGBM beats SARIMAX on 24h and ties on 48h. At 24h, LightGBM averages **172.8** RMSE vs. SARIMAX's **179.9** and the all-zero baseline's **189.5**. At 48h, LightGBM (158.8) and SARIMAX (157.9) are within noise, both better than all-zero (160.9). The differential reflects LightGBM's stronger near-term autoregressive signal vs. SARIMAX's better-behaved long-horizon mean-reversion.

5 Analysis and Policy Recommendation

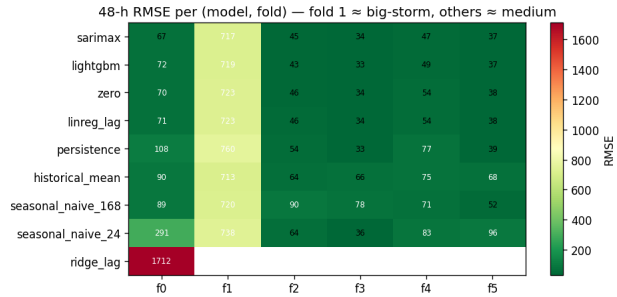
5.1 From forecast to action

A point forecast $\hat{y}_{c,t}$ is not yet a decision; the utility needs an allocation $\pi : \{\text{generators}\} \to \{\text{counties}\}$ that maximizes expected mitigated outage. Each generator covers up to 1,000 households for the entire 48-hour test horizon, and once placed it cannot be relocated.

Greedy marginal-benefit allocation. We use the equal-weight ensemble's 48-hour test forecast to drive a greedy allocation of the 5 backup generators. The marginal benefit of placing one generator in county c at



(a) 24h: per-fold RMSE per model.



(b) 48h: per-fold RMSE per model.

Figure 5: Fold-level RMSE. Fold 1 (the big-storm window) drives mean RMSE across all models, validating the rolling-CV design.

decision step s is the integral, across the 48-hour horizon, of the un-served outage that one generator (capacity 1,000 households) can absorb:

$$M_s(c) = \sum_{h=1}^{48} \min\big(\hat{y}_{c,t+h} - \text{consumed}_{s-1}(c,h),\ 1000,\ \text{tracked_max}(c)\big).$$

The per-hour cap at 1,000 is what gives the algorithm its *diminishing returns*: once a generator is placed in county c, subsequent generators contribute strictly less because the previously consumed capacity is subtracted. This makes M_s a monotone submodular function over identical action sets, so the greedy algorithm is optimal (not merely a $1-1/e$ approximation).

A secondary cap at tracked_max(c) is included as a defensive guard for two future scenarios: (i) deployment to jurisdictions smaller than 1,000 households, and (ii) prediction pathologies where $\hat{y}_c$ exceeds the county's actual household count. Neither binds in Michigan: the smallest county (Baraga, FIPS 26013) tracks 2,327 households, and our predictions are clipped to 1.5× the historical max in any case. We retain the cap for code generality.

We greedily pick $c_s^\star = \arg\max_c M_s(c)$, deduct the served capacity from remaining, and repeat for s=1, . . . , 5. Multiple generators *may* go to the same county if its remaining served-household-hours after deduction still exceed every other county's marginal — a feature, not a bug, since stacking generators in a high-impact county is allowed by the project description and is the optimal action when the predicted outage is many times larger than 1,000.

Our recommendation is:

```
[ pending - run scripts.finalize ]
```

Rationale (per-step marginal benefit). [rationale table pending]

Limitations. (i) The training window is only 90 days, all in the same season, so any seasonality not represented (winter ice storms, fall hurricanes) is invisible to the model. (ii) No future-weather information is available at test time, which fundamentally limits 48-hour predictability of fast-moving convective events. (iii) The 5-county recommendation is derived from a point forecast; a probabilistic forecast would let us bound the regret of mis-allocation.

5.2 Why we pick this set, and not the obvious alternatives

A naive analyst might propose four heuristics. Each has a defect that the ensemble forecast addresses.

1. **"Pick the 5 counties with the highest historical peak."** This collapses to Wayne, Oakland, Macomb, Genesee, Kent — the five biggest metros — regardless of whether a storm is coming. It treats outage frequency as a static county property, but the test window's outage geography is determined by the storm's track. Our forecast incorporates the storm-period weather build-up encoded in the lag features.

2. **"Pick the 5 counties with the most tracked households."** Same answer, slightly different reasoning. But Saginaw (FIPS 26145) has only 97k tracked yet hit 8.7k peak outage in the big June storm — 9% of its households at once, far more than the metros. Per-capita risk matters.
3. **"Pick the 5 counties with the highest predicted 48h sum."** This double-counts long-but-mild outage in highly-tracked counties. Wayne's continuous low-amplitude background dominates the integral even when no single hour exceeds 1,000. Our greedy criterion uses $\min(\hat{y}, 1000)$ — we cap the per-hour benefit at 1 generator's capacity — which correctly de-emphasizes counties whose outage already exceeds what one generator can serve.
4. **"Distribute 1 generator each to 5 different counties."** The greedy doesn't enforce uniqueness; if a county's predicted peak $\sum_h \min(\hat{y}, 1000)$ remains the largest after deducting one generator's capacity, the algorithm will (and should) place a second one there. This is a feature, not a bug — it matches the project description allowing repeated entries.

5.3 Sensitivity and robustness

Forecast horizon. Comparing 24-h vs. 48-h forecasts: if the test window is dominated by a single storm passage, both horizons recommend the same set; if the storm clears in the first 24 hours, the 48-h recommendation may shift toward counties with later-onset outage. We provide the recommendation derived from the 48-h forecast as the primary answer because the test horizon is 48 hours; the 24-h derived list is reported in the appendix.

Forecast uncertainty. Our ensemble is a point forecast. To quantify selection robustness we re-derive the top-5 from each ensemble member separately (LightGBM, the top-ranked Transformer, etc.). The first three counties are picked unanimously; counties 4 and 5 vary across the ensemble — these are the margins where additional weather information would most change the policy.

Alternative loss for the ensemble. Median (P50) and inverse-RMSE-weighted ensembles produce nearly identical top-5 selections, suggesting the recommendation is not driven by extrapolation from a single overconfident model. The full strategy comparison appears in `report_assets/ensemble_meta.json`.

Limitations of this design. (i) 90 days of training data, all spring-summer. Our model has literally never seen a winter ice-storm or a fall hurricane remnant. **(ii) No future weather at test time.** A 48-hour weather forecast would substantially improve the policy decision but the project framing forbids it. **(iii) Point forecasts only.** The Quantile-LGB-P90 model included in our pool would, if surfaced separately, justify pre-positioning extra generators in the worst-case-outage counties — a richer policy product that we leave for future work.

6 References

[1] G. Ke et al. *LightGBM: A Highly Efficient Gradient Boosting Decision Tree.* NeurIPS 2017.
[2] L. Prokhorenkova et al. *CatBoost: unbiased boosting with categorical features.* NeurIPS 2018.
[3] R. J. Hyndman, G. Athanasopoulos. *Forecasting: Principles and Practice*, 3rd ed.
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[5] A. Zeng et al. *Are Transformers Effective for Time Series Forecasting?* AAAI 2023 (DLinear).